

Benchmark & Dataset Paper Reading Template

- Purpose: Critically read, analyze, and discuss benchmark & dataset papers
- Focus: Reasoning, design quality, limitations, and rebuttal analysis

Task & Problem Formulation

- What is the learning task?
- Is the task well-defined and realistic?
- Assumptions made (explicit & implicit)
- Alignment with real-world use cases

Dataset Construction

- Data source and collection process
- Scale and diversity
- Biases, noise, missing data
- Train/validation/test splits

Benchmark Design

- What is being benchmarked and why?
- Baselines: are they strong and appropriate?
- Evaluation metrics: do they match the task?
- Risk of leaderboard overfitting

Experimental Protocol

- Reproducibility and clarity
- Hyperparameter fairness
- Leakage or shortcuts?
- Are results statistically meaningful?

Strengths

- What is genuinely new or valuable?
- What problems does this benchmark solve well?
- Community impact

Weaknesses & Limitations

- Design flaws or missing cases
- Unrealistic assumptions
- Dataset bias or artifacts
- What could break this benchmark?

Comparison to Prior Work

- How does this differ from existing benchmarks?
- Is it incremental or transformative?
- Fairness of comparisons

Ethics & Responsible Use

- Potential misuse or harm
- Privacy and consent
- Dataset governance

OpenReview Reviews – Summary

- Main reviewer concerns
- Repeated criticisms
- Score distribution

Rebuttal Analysis

- Do authors address concerns clearly?
- Are responses technical or evasive?
- Which critiques remain unresolved?

Critical Reasoning Exercise

- If you were a reviewer: accept or reject?
- What would you request for revision?
- What experiments are missing?

Main Points of Critique & Rebuttal

1. Critique: **Not entirely novel**

“Most of the previous works focus on neural signals recorded from EEG and fMRI where the temporal dynamics could be different from the field potential (Local Field Potential and ECoG) datasets that we study in this work. In addition, most of the previous works address the classification task. Whereas we focus on the forecasting task ...”

2. Critique: **Major issue in the experimental design by split the data train and test sets only**

“We further investigated this point by validating AMAG and GNN-based baselines with a train-val-test split... In Table D1, we report the performance on the new test set from this train-val-test split ...”

3. Critique: **It is unclear where the predictive improvements are coming from**

“We included new variants in the ablation study (for the SI block) ...”

4. Critique: **Further investigation on model complexity and scalability**

“AMAG generally requires fewer parameters and less training time (compared to graph-based methods). The major bottleneck of AMAG is the requirement for larger memory, thus, scaling AMAG could be limited by the availability of working memory.”

5. Critique: **The results provide limited insight to verify the hypothesis**

“We additionally visualized the learned adjacency matrices for both the Add module (Aa) and the Modulator module (Am) when AMAG is trained to perform one-step forecasting and multi-step forecasting ... each has distinct local patterns. Am appears to be denser, while Aa is sparser. Looking at specific target channels, some channels appear to contribute more in the additive module, while other channels could be more important in the modulator module”

Design Thinking & Re-evaluation



Approach taken by
the Authors

Step

Description

Scoring

Problem
Identification

- Integrates Multimodal ML with deep GNNs for learning unified gene representations from multimodal biological data
- Highly relevant in computational biology

- Clarity of problem definition: 7/10

Solution
Design

- Effectively combines multiple techniques to learn gene representations containing functional similarity across context in the joint latent space

- Improvement upon existing methods: 7/10
- Clarity of design principles and methodologies: 5/10

Evaluation
Methods

- Compared across:
- 9 baselines, both DL and non-DL based
 - Different tissues and biological downstream tasks

- Thoroughness and Persuasiveness: 7/10
- Alignment with stated objectives: 7/10

Lessons Learnt



Key Learnings

- Very strong and extensive empirical analysis required for specialised domains
- Papers with limited methodological novelty is only accepted if
 - The **problem** they are solving **is highly relevant**
 - They **lead to significant improvement** against baselines
- Techniques to **pivot discussion** and how to **construct strong counter-arguments**